

Shuigen Liu

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Positions

2026 Aug – **Postdoctoral Fellow**, The Pennsylvania State University
2025 – 2026 **Research Fellow**, National University of Singapore
2024 Jul – Dec **Visiting student**, Massachusetts Institute of Technology

Education

2021 – 2025 **Ph.D. in Mathematics**, National University of Singapore
Advisors: Prof. Xin T. Tong and Prof. Weizhu Bao
2017 – 2021 **B.S. in Mathematics**, Peking University

Research Interests

My research lies at the intersection of computational mathematics, applied probability and statistics. I focus on analysis and development of algorithms for high-dimensional sampling and uncertainty quantification, with applications in Bayesian inference, inverse problems and generative modeling. I am particularly interested in localization methods for sampling. I also work on ensemble Kalman methods for high-dimensional inference and data assimilation.

Keywords: Sampling, uncertainty quantification, localization, dimension reduction, data assimilation.

Publications and Preprints

Preprints

- [6] Georg A. Gottwald, **Shuigen Liu**, Youssef Marzouk, Sebastian Reich, and Xin T. Tong, *Localized diffusion models*, arXiv:2505.04417, submitted to Bernoulli (2026).
[5] Tiangan Cui, **Shuigen Liu**, and Xin T. Tong, *Stein's method for marginals on large graphical models*, arXiv:2410.11771, submitted to Bernoulli (2026).

Published/accepted

- [4] **Shuigen Liu**, and Xin T. Tong, *Spectral Gap of Metropolis Algorithms for Non-smooth Distributions under Isoperimetry*, Jpn. J. Stat. Data Sci. (2026).
[3] Ningxin Liu, **Shuigen Liu**, Xin T. Tong, and Lijian Jiang, *Estimate of Koopman modes and eigenvalues with Kalman filter*, SIAM J. Sci. Comput., 48 (2026), pp. A512-A539.
[2] Rafael Flock, **Shuigen Liu**, Yiqiu Dong, and Xin T. Tong, *Local MALA-within-Gibbs for Bayesian image deblurring with total variation prior*, SIAM J. Sci. Comput., 47 (2025), pp. A2127-A2153.
[1] **Shuigen Liu**, Sebastian Reich and Xin T. Tong, *Dropout ensemble Kalman inversion for high dimensional inverse problems*, SIAM J. Numer. Anal., 63 (2025), pp. 685-715.

In preparation

Shuigen Liu, Youssef Marzouk, Aimee Maurais, and Xin T. Tong, *Sampling by Localized Flows*, (2026+).

Tiangang Cui, **Shuigen Liu**, Xin T. Tong, Romain Verdiere, and Olivier Zahm *Localized Triangular Transport Map*, (2026+).

Teaching

National University of Singapore

- TA Finite Element Method, Spring 2025
- TA Essential Data Analytics Tools: Numerical Computation, Spring 2024
- TA Calculus for Computing, Spring 2023 and Fall 2023
- TA Essential Data Analytics Tools: Convex Optimization, Spring 2023

Awards

- 2024** Faculty of Science PhD Conference Award, National University of Singapore
- 2024** Overseas Research Immersion Award, National University of Singapore
- 2021** President's Graduate Fellowship, National University of Singapore
- 2021** Excellent Graduate, Peking University

Service

Reviewer for: SIAM Journal on Numerical Analysis, SIAM/ASA Journal on Uncertainty Quantification, Journal of Machine Learning Research, Statistics and Computing, Japanese Journal of Statistics and Data Science

Conferences

- Jul 2025** Efficient Sampling Algorithms for Complex Models, IMS NUS, Singapore.
- Jun 2025** MATRIX Research Program: Bayesian Learning of Very High-Dimensional Physical Process Models, MATRIX, Melbourne, Australia.
- May 2025** 2025 STU-NUS PhD Forum, Shanghai Jiao Tong University, Shanghai, China.
- Oct 2024** FDS Conference: Recent Advances and Future Directions for Sampling, Yale University, New Haven, U.S.
- Feb 2024** 2024 SIAM Conference on Uncertainty Quantification (UQ24), Trieste, Italy.
- Aug 2023** 2023 International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan.
- Jun 2023** Foundations of Computational Mathematics 2023 Conference (FoCM 2023), Sorbonne University, Paris, France.
- Jul 2022** 2022 International Conference on Scientific Computation and Differential Equations (SciCADE 2022), University of Iceland, Reykjavík, Iceland.